

DS 6021 Intro to Predictive Modeling Lab 4 Part 3: Final Project Update

TEAM

Fiona St.Clair (nye2xd), Hannah Sweazey (qcp6vj), Kim Mondragon (phy2zm), Shane Brosnan (atu4xv), and Lior Belay (qyc6ej).

BASIC PROPOSAL

Our team is exploring ticket pricing for live music events. Ticket prices for the same artist can vary widely across venues and markets, and it's not always clear how much of that variation reflects genuine differences in demand versus other factors like venue size, location, or timing. We plan to combine data from the Ticketmaster Discovery API (event listings, price ranges, venue details, genre, and event dates) with the Spotify Web API (artist popularity scores and follower counts, which we're using as a proxy for audience demand), and potentially Census demographic data by metro area to capture local market effects. Our core problem is to build a predictive model of ticket price using these features, and then use the model's residuals to identify which artists or shows are priced notably above or below what their popularity and market would predict, essentially flagging mispricing rather than just forecasting a number. We'll pair this with clustering to see whether artists group into natural pricing "tiers" that aren't fully explained by popularity alone.

NOTES

- Need to narrow down exactly with features we select. We don't want to do have too many features that complicates the model beyond what we need.
- Potential time component piece...TBD.